

Q1: Verify Gauss' Divergence Theorem for the vector field $\vec{F} = (y, 2x, -z)$ in the region defined by $0 \leq z \leq 4$, $0 \leq y$, and $2x + y \leq 6$.

The Divergence Theorem of Gauss:

$$\int \nabla \cdot \vec{F} dV = \int \vec{F} \cdot \hat{n} dS$$

Vector Field:

$$\vec{F} = (y, 2x, -z) = y\hat{e}_1 + 2x\hat{e}_2 - z\hat{e}_3$$

For $\int \nabla \cdot \vec{F} dV$:

$$\int \nabla \cdot \vec{F} dV = -36$$

For $\int \vec{F} \cdot \hat{n} dS$:

For diagonal plane $((V) : \int \vec{F} \cdot \hat{n} dS)$ use plane equation:

$$\int \vec{F} \cdot \hat{n} dS = -36$$

$$\int \nabla \cdot \vec{F} dV = \int \vec{F} \cdot \hat{n} dS = -36$$

Q2: For a volume V and bounded surface S with the outward unit vector $\hat{\mathbf{n}}$, if $\mathbf{x}$ is the position vector of any point in the domain, prove the identities below.

$$a. \int_S \nabla(\mathbf{x} \cdot \mathbf{x}) \cdot \hat{\mathbf{n}} dS = 6V$$

$$b. \int_S \mu(\mathbf{x}) \operatorname{curl}(\mathbf{v}) \cdot \hat{\mathbf{n}} dS = \int_V \operatorname{curl}(\mathbf{v}) \cdot \operatorname{grad}(\mu(\mathbf{x})) dV$$

$$a. \int_S \nabla(\mathbf{x} \cdot \mathbf{x}) \cdot \hat{\mathbf{n}} dS = 6V$$

Gauss Theorem:

$$b. \int_S \mu(\mathbf{x}) \operatorname{curl}(\mathbf{v}) \cdot \hat{\mathbf{n}} \, dS = \int_V \operatorname{curl}(\mathbf{v}) \cdot \operatorname{grad}(\mu(\mathbf{x})) \, dV$$

Q3: For the position vector $\mathbf{x}$ having the magnitude x , show that:

$$\mathbf{a.} \, x_{,j} = \frac{x_j}{x}, \quad \mathbf{b.} \, (x^{-1})_{,ij} = \frac{3x_i x_j}{x^5} - \frac{\delta_{ij}}{x^3}$$

a. $x_{,j} = \frac{x_j}{x}$

b. $(x^{-1})_{,ij} = \frac{3x_i x_j}{x^5} - \frac{\delta_{ij}}{x^3}$

Q4: For the position vector $\mathbf{x}$ in a closed region R , show that

$$\int_R [\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi] dV = \int_S \phi \frac{\partial \psi}{\partial n} dS$$

Use Gauss Theorem:

Q5: Use $\nabla \phi = \lim_{\nabla V \rightarrow 0} \frac{1}{\nabla V} \int_S \mathbf{n} \phi dS$ and find the nabla operator in cylindrical coordinate system.

Control Volume:

$$r^\pm = r \pm \frac{dr}{2}, \quad \theta^\pm = \theta \pm \frac{d\theta}{2}, \quad z^\pm = z \pm \frac{dz}{2}$$

$$\Delta V = (dr)(rd\theta)(dz) = r dr d\theta dz, \quad \phi_0 = \phi(r, \theta, z)$$

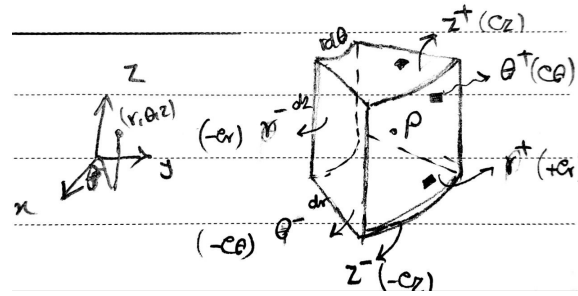


Figure 1: Cylindrical element

Position	dS	$\hat{n}$	ϕ
r^+	$(r + \frac{dr}{2})d\theta dz$	$\hat{e}_r$	$\phi_0 + \frac{dr}{2} \frac{\partial \phi}{\partial r}$
r^-	$(r - \frac{dr}{2})d\theta dz$	$-\hat{e}_r$	$\phi_0 - \frac{dr}{2} \frac{\partial \phi}{\partial r}$
θ^+	$drd\theta dz$	$\hat{e}_\theta^+$	$\phi_0 + \frac{d\theta}{2} \frac{\partial \phi}{\partial \theta}$
θ^-	$drd\theta dz$	$-\hat{e}_\theta^-$	$\phi_0 - \frac{d\theta}{2} \frac{\partial \phi}{\partial \theta}$
z^+	$rd\theta dr$	$\hat{e}_z$	$\phi_0 + \frac{dz}{2} \frac{\partial \phi}{\partial z}$
z^-	$rd\theta dr$	$-\hat{e}_z$	$\phi_0 - \frac{dz}{2} \frac{\partial \phi}{\partial z}$

Table 1: Gauss’s theorem terms

$$\nabla = \hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{e}_\psi \frac{\partial}{\partial \psi}$$

Q6: Derive nabla operator for spherical coordinate system. Then find $\vec{\nabla} \times \vec{V}$ in spherical coordinate system.

$$r^\pm = r \pm \frac{dr}{2}, \quad \theta^\pm = \theta \pm \frac{d\theta}{2}, \quad \psi^\pm = \psi \pm \frac{d\psi}{2}$$

$$\Delta V = dr(r d\theta)(r \sin \theta d\psi) = r^2 \sin \theta dr d\theta d\psi, \quad \phi_0 = \phi(r, \theta, \psi)$$

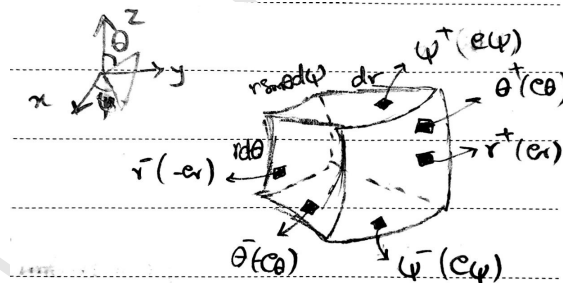


Figure 2: Spherical element

Table 2: Gauss’s theorem terms

[Redacted area]

$$\nabla = \hat{e}_r \phi_{,r} + \hat{e}_\theta \frac{1}{r} \frac{\partial \phi}{\partial \theta} + \hat{e}_\psi \frac{1}{r \sin \theta} \frac{\partial \phi}{\partial \psi}$$

for $\vec{V} = V_r \hat{e}_r + V_\theta \hat{e}_\theta + V_\psi \hat{e}_\psi$:

[Redacted area]

$$\vec{a} = a_r \hat{e}_r + a_\theta \hat{e}_\theta + a_z \hat{e}_z$$

$$a_r = \frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} v_\theta + v_z \frac{\partial v_r}{\partial z}$$

$$a_\theta = \frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} v_r + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z}$$

$$a_z = \frac{\partial v_z}{\partial t} + v_z \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z}$$

Q8: Determine the principal values and principal directions of the matrix representation of tensor T .

$$T_{ij} = \begin{bmatrix} 30 & 13 & -21 \\ 13 & 42 & 34 \\ -21 & 34 & 55 \end{bmatrix}$$

Use $T.n = \lambda.n$:

$$\Rightarrow \lambda_1 = 84.45, \lambda_2 = 45.60, \lambda_3 = -3.05, \text{ for normalize: } \hat{n}_i^k \cdot \hat{n}_j^k = 1$$

for λ_1 :

$$\rightarrow \begin{bmatrix} n_1^{(1)} \\ n_2^{(1)} \\ n_3^{(1)} \end{bmatrix} = \begin{bmatrix} -0.1665 \\ 0.5848 \\ 0.7938 \end{bmatrix}$$

for λ_2 :

$$\rightarrow \begin{bmatrix} n_1^{(2)} \\ n_2^{(2)} \\ n_3^{(2)} \end{bmatrix} = \begin{bmatrix} 0.7933 \\ 0.5575 \\ -0.2442 \end{bmatrix}$$

for λ_3 :

$$\sqrt{C} = \begin{bmatrix} 5.3540 & -1.8343 & 0.9831 \\ -1.8343 & 4.2658 & 0.6595 \\ 0.9831 & 0.6595 & 6.0493 \end{bmatrix}$$

Q10: Rewrite the following equations using indicial notation.

a. $a_1x_1x_3 + a_2x_2x_3 + a_3x_3x_3$

$$\text{[Blank Box]} = a_i x_i x_3$$

b.
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases}$$

$$\text{[Blank Box]} \Rightarrow a_{ij}x_j = b_i$$

Q11: Prove that any isotropic third order tensor T_{ijk} may be presented as $T_{ijk} = \lambda \epsilon_{ijk}$ where λ is scalar

